

Dhruv Rastogi

AI / Backend Engineer — LLM Applications · RAG · Distributed Systems

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SUMMARY

Software engineer with around 15 months of experience building across the AI application stack, from backend services in Java and Spring Boot to LLM-powered products. Built AI-Court, a deployed Retrieval-Augmented Generation (RAG) platform on LangChain with citation-grounded answers for verifiability, and DhrLang, a statically-typed JVM language with a full compiler and three execution backends (1,000+ tests in CI). Currently at MAQ Software on 500K+ records-a-day Azure data pipelines. Comfortable across LLM application design, retrieval systems, Python, and production backend engineering.

SKILLS

AI / LLM: LLM application development, Retrieval-Augmented Generation (RAG), LangChain, prompt design, retrieval pipelines, citation-grounded UX

Languages: Python, Java (primary), JavaScript / TypeScript, SQL, Bash

Backend: Spring Boot, REST APIs, microservices, authentication, role-based access control (RBAC), async / queue-based processing

Cloud & Data: Microsoft Azure (Functions, Fabric, Databricks, DevOps), Docker, Kubernetes, Terraform, MySQL, PostgreSQL, MongoDB, Redis, ETL pipelines

Compilers & Testing: Compiler design (lexer / parser / IR / JVM bytecode), Language Server Protocol (LSP), JUnit, PIT mutation testing, JaCoCo

PROJECTS

AI-Court — RAG legal-assistance platform github.com/dhruv-15-03/AI-CourtRoom

Two-tier Retrieval-Augmented Generation app on LangChain (Python core with a JavaScript shell), deployed on Vercel. Every answer links back to its source documents, so wrong responses are traceable instead of a black box.

DhrLang — statically-typed JVM language (v3.0.0) github.com/dhruv-15-03/DhrLang

Compiler written end to end: lexer, parser, type checker, IR lowering, and three execution backends (AST interpreter, IR VM, JVM bytecode), plus an EVM target. Ships an LSP server with JSON diagnostics and a published VS Code extension. 14 releases; 1,000+ tests in CI with JaCoCo coverage and PIT mutation testing.

boot-usage — Spring Boot Actuator starter (Apache-2.0) github.com/dhruv-15-03/boot-usage

Drop-in Spring Boot starter that exposes Actuator endpoints for runtime dependency analysis. Published as a reusable Java package.

EXPERIENCE

Associate Software Engineer · MAQ Software Nov 2025 – Present · India

Java · Python · Azure Fabric · Databricks · Azure Functions · Terraform · Azure DevOps

- Build and run ETL pipelines on Azure Fabric and Databricks that process 500,000+ records a day at 99.5% accuracy, cutting manual data prep by about 60%.
- Own CI/CD for 10+ Azure Function Apps using Azure DevOps and Terraform, enabling zero-downtime releases and roughly 40% faster deployments.
- Improved alerting, monitoring, and reliability tuning for services running on a 99.9% uptime SLA.

Associate Software Engineer (Contract) · RecruitCRM Aug 2025 – Oct 2025 · Remote

Java · Spring Boot · MySQL · REST APIs · Vue.js · RBAC

- Shipped 8+ REST APIs for CRM workflows, lead enrichment, and live data access, with 85% test coverage on the new code.
- Implemented authentication and role-based access control across the backend.
- Reduced API response times by 30% through query and execution-path optimization.

Software Engineer Intern · CEERAS Feb 2025 – Jun 2025 · Remote

Java · Microservices · Docker · Kubernetes · Redis · GitHub Actions

- Built a distributed bidding system handling 10,000+ transactions a day at sub-second latency.
- Added asynchronous processing and a Redis cache layer to raise end-to-end throughput.
- Set up the Docker, Kubernetes, and GitHub Actions pipeline for build, test, and deployment.

EDUCATION

B.Tech, Computer Science & Engineering (AI Specialization) Expected Jul 2026

Dr. A.P.J. Abdul Kalam Technical University (AKTU), India · CGPA 8.5 / 10 · Pursued while working full-time.

CERTIFICATIONS

Microsoft Certified — [SQL AI Developer Associate](#) · [Azure Developer Associate, AZ-204](#) · [DevOps Engineer Expert, AZ-400](#) · Fabric Analytics Engineer, DP-600

Oracle Cloud Infrastructure 2025 — [Developer Professional](#) · [DevOps Professional](#)

[MongoDB Associate Developer, Java](#) · [GitHub Copilot Certified](#) · Full verified list: dhruvrastogi.me/certifications

ACHIEVEMENTS

LeetCode Knight (1,000+ problems solved) · Smart India Hackathon 2024 — Top 50 nationally · ISC Class XII — 96.75% (Rank 1 on campus)

Available in 2 weeks. Open to AI Engineer, Applied AI / LLM, and Backend roles — remote, hybrid, or relocation.